VF Asia Footwear Laboratory Testing Manual

Test Method for Weight	Test method #: VF-A02 Page 1 of 2
Date Issued: November 20, 2018 Revision #: 1 Reason for Revisions: 1. Change measuring tool to caliper for better accuracy > 2. Change dimension of test sample > 3. Increase quantity of samples to 3.	Created by: Nila Babia Erazo

PURPOSE AND SCOPE: To measure the mass per unit area (or mass per length) of a test material. This is applicable for all solid materials.

REFERENCE: ASTM D3776

EQUIPMENT AND MATERIAL:

1. Balance capable of weighing to 0.001 grams.
2. Digital calipers capable of measuring up to 0.01mm.
3. Leather, synthetic leather, fabric, thread, shoe laces, and paper / fiber board.

SAMPLE PREPARATION:

Condition the sample for 24 hours at standard laboratory environment @ 23.0±2.0 ˚C Temperature and 55 ±10 % Relative Humidity.

A) Leather, synthetic leather, fabric, and paper/fiber board.
Die cut a square test sample with sides of 10.0 cm.

B) Shoe lace
Cut a sample 1.00 ±0.05 meters long. The aglets must be cut off.

C) Thread
Cut a sample 5.00 ±0.05 meters long.

REPETITIONS: Test three (3) samples per material.

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.

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1. Calibrate the balance according to the operating instructions of the equipment being used and /or the **VF Asia** Calibration Manual.
2. • Using the digital caliper, measure the dimensions of the sample and calculate the area of the sample to the nearest 0.1cm^2(**A**).
   • *For thread and shoe lace samples, measure the length to the nearest 0.1cm (**L**).*
3. Place the test sample on the balance. Do not allow any portion of the test sample to overhang the perimeter of the weighing pan.
4. Record the weight of the sample to the nearest 0.001g (**W**).

CALCULATIONS:

1. **Sample A:**

$$\text{Weight per unit area (g/m}^2\text{)} = \frac{W}{A} \times 10000$$

2. **Sample B or C:**

$$\text{Weight per unit length (g/m)} = \frac{W}{L} \times 100$$

REPORT: The average of the 3 measurements.